

Section : Chemistry

Q.1 Geometry of  $[\text{Ni}(\text{CN})_4]^{2-}$  and  $[\text{NiCl}_4]^{2-}$  are

1. Tetrahedral, low spin and square planar, high spin respectively
2. Tetrahedral, high spin and square planar, low spin respectively
3. Square planar, low spin and tetrahedral, high spin respectively
4. Square planar, high spin and tetrahedral, low spin respectively

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Specific conductance ( $\kappa$ ) is related to resistivity as

1.  $\kappa = R\rho$

2.  $\kappa = \frac{1}{R} \times \rho$

3.  $\kappa = \frac{R}{\rho}$

4.  $\kappa = \frac{1}{\rho}$

Options 1. 1

2. 2

3. 3

4. 4

Q.3 Lactose on hydrolysis gives

1.  $\beta$ -D-Galactose and  $\beta$ -D-Glucose
2.  $\alpha$ -D-Galactose and  $\beta$ -D-Glucose
3.  $\alpha$ -D-Glucose and  $\beta$ -D-Glucose
4.  $\alpha$ -D-Galactose and  $\alpha$ -D-Glucose

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Match the properties given in List-I with the metals given in List-II

List-I

List-II

(Property)

(Metal)

(A) An element which can show +8 oxidation state

(I) V

(B) 3d block element that can show upto +7 oxidation state

(II) Cr

(C) 3d block element with the highest melting point

(III) Os

(D) 3d block element with the highest enthalpy of atomization

(IV) Mn

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.5 Match List-I (vitamins) with List-II (Deficiency diseases)

List-I

List-II

Name of vitamins Deficiency disease

(A) Vitamin C

(I) Night blindness

(B) Vitamin K

(II) Rickets

(C) Vitamin D

(III) Increased blood clotting time

(D) Vitamin A

(IV) Scurvy

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.6 The major product formed when acetanilide undergoes bromination in presence of acetic acid is

1. 2,4,6-Tribromoacetanilide
2. m-bromoacetanilide

3. p-bromoacetanilide  
4. o-bromoacetanilide

Options 1. 1

2. 2  
3. 3  
4. 4

**Q.7** Curium (Cm) belongs to 5f series. Its atomic number is 96. Which of the following is the correct electronic configuration of Curium?

1.  $[\text{Rn}] 5f^7 6d^1 7s^2$   
2.  $[\text{Rn}] 5f^7 6d^3$   
3.  $[\text{Rn}] 5f^7 6d^2 7s^1$   
4.  $[\text{Rn}] 5f^{10}$

Options 1. 1

2. 2  
3. 3  
4. 4

**Q.8** For a chemical reaction.

- (A) The order of a reaction is the sum of powers of the concentration of the reactants.  
(B) The order of a reaction may be zero, an integer or a fractional value.  
(C) Molecularity of an elementary reaction is equal to the number of reacting species.  
(D) The molecularity of a reaction may be zero or a fractional value.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only  
2. (D), (B) and (C) only  
3. (A), (B), (C) and (D)  
4. (A), (B) and (C) only

Options 1. 1

2. 2  
3. 3  
4. 4

**Q.9** Gabriel synthesis is used for the synthesis of

1. Carboxylic acids  
2. Primary amines  
3. Alcohols  
4. Aldehydes

Options 1. 1

2. 2  
3. 3  
4. 4

**Q.10** The rate constant, activation energy and the Arrhenius parameter at  $27^\circ\text{C}$  are  $2.0 \times 10^{-5} \text{ s}^{-1}$ ,  $100 \text{ kJ mol}^{-1}$  and  $6.0 \times 10^{14} \text{ s}^{-1}$  respectively. The value of the rate constant at a very high temperature ( $T \rightarrow \infty$ ) approaches to:

1.  $2.0 \times 10^{-5} \text{ s}^{-1}$   
2. Infinite  
3.  $6.0 \times 10^{14} \text{ s}^{-1}$   
4.  $4 \times 10^{-5} \text{ s}^{-1}$

Options 1. 1

2. 2  
3. 3  
4. 4

**Q.11** Match List-I with List-II :

List-I  
(Items)

List-II

(Properties)

- (A) Primary batteries (I) Pollution free energy converter.  
(B) Secondary batteries (II) Takes place in the presence of air, moisture and  $\text{H}^+$   
(C) Fuel cells (III) Large number of charging/discharging and long-lasting  
(D) Corrosion (IV) Mercury cell is used in hearing aids

Choose the correct answer from the options given below:

1. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)  
2. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)  
3. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)  
4. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Options 1. 1

2. 2  
3. 3  
4. 4

**Q.12** Arrange the following alkyl halides in increasing order of reactivity towards  $\text{S}_\text{N}^2$  reaction?

- (A)  $\text{CH}_3\text{-Cl}$

(B)  $\text{CH}_3\text{-Br}$

(C)  $\text{CH}_3\text{-F}$

(D)  $\text{CH}_3\text{-I}$

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (B), (D), (C)
3. (B), (A), (D), (C)
4. (C), (A), (B), (D)

Options 1. 1

2. 2
3. 3
4. 4

**Q.13** Aryl Halides are less reactive towards nucleophilic substitution reactions because of

- (A) Resonance Effect
- (B) Stability of phenyl cation
- (C) Presence of  $\text{sp}^2$  Carbon attached to halogen
- (D) Repulsion between benzene ring and incoming nucleophile

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

**Q.14** Arrange the following complexes in increasing order of conductivity of their solution:

- (A)  $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- (B)  $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
- (C)  $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (D)  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

Choose the correct answer from the options given below:

1. (A), (B), (D), (C)
2. (B), (C), (A), (D)
3. (B), (C), (D), (A)
4. (C), (D), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

**Q.15** Which of the following material is a semiconductor?

1. Silver
2. Silicon
3. Teflon
4. Glass

Options 1. 1

2. 2
3. 3
4. 4

**Q.16** Fructose is a ketohexose. The keto group is positioned at which carbon

1. Carbon 2
2. Carbon 3
3. Carbon 4
4. Carbon 5

Options 1. 1

2. 2
3. 3
4. 4

**Q.17** The standard reduction potential ( $E^\circ$ ) values of  $\text{Ag}^+|\text{Ag}$ ,  $\text{K}^+|\text{K}$ ,  $\text{Mg}^{2+}|\text{Mg}$  and  $\text{Cr}^{3+}|\text{Cr}$  are 0.80 V, - 2.93 V, - 2.37 V and - 0.74 V respectively. The decreasing order of reducing power of the following metals is

- (A) Ag
- (B) K
- (C) Mg
- (D) Cr

Choose the correct answer from the options given below:

1. (A), (D), (C), (B)
2. (A), (B), (C), (D)
3. (B), (C), (D), (A)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.18 The halide which will undergo  $\text{SN}^2$  reaction easily

1. Benzyl Chloride.
2. Ethyl Chloride.
3. Tertiary Butyl Chloride.
4. Allyl Chloride

Options 1. 1

2. 2

3. 3

4. 4

Q.19  $\Lambda_m^\circ$  values for electrolytes AB, HB and AD are 120.5, 300.9 and  $80.0 \text{ S cm}^2 \text{ mol}^{-1}$  respectively.

$\Lambda_m^\circ$  value for HD is

1.  $260.4 \text{ S cm}^2 \text{ mol}^{-1}$
2.  $470.4 \text{ S cm}^2 \text{ mol}^{-1}$
3.  $340.4 \text{ S cm}^2 \text{ mol}^{-1}$
4.  $100.4 \text{ S cm}^2 \text{ mol}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.20 The composition of Fehling solution A and Fehling solution B respectively are

1. The aqueous solution sodium potassium tartarate and copper sulphate respectively
2. The aqueous solution of copper sulphate and acetic acid respectively
3. The aqueous solution of copper sulphate and formic acid respectively
4. The aqueous solution of copper sulphate and alkaline sodium potassium tartarate respectively

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Which of the following compounds reacts with acids to give salts?

1. Ethers
2. Phenols
3. Amines
4. Carbonyl Compound

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Which of the following reactions is used to prepare Haloarenes ?

1. Stephen reaction
2. Gatterman-Koch reaction
3. Cannizzaro reaction
4. Sandmeyer reaction

Options 1. 1

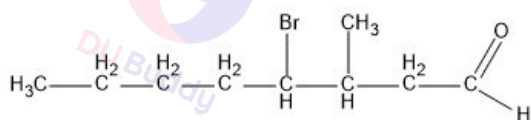
2. 2

3. 3

4. 4

Q.23

The correct IUPAC name for the following structure is



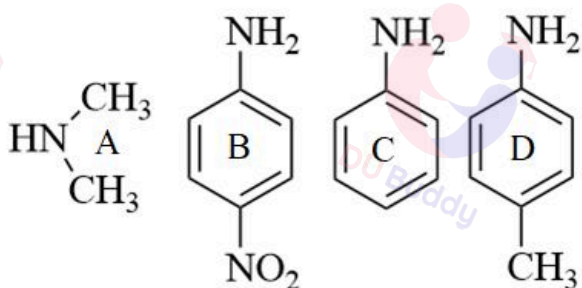
1. 4-Bromo-3-methylheptanol
2. 4-Bromo-5-methylheptanal
3. 4-Bromo-5-methylheptanol
4. 4-Bromo-3-methylheptanal

Options 1. 1

2. 2

Q.24

Arrange the following amines in order of their increasing basic strength



1. (A), (C), (D), (B)
2. (C), (A), (D), (B)
3. (B), (C), (D), (A)
4. (D), (C), (B), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Which of following statement(s) are correct for Glucose ?

- (A) Glucose is a monosaccharide sugar  
 (B) Glucose is an aldohexose and is also known as dextrose.  
 (C) Glucose can be hydrolyzed.  
 (D) Glucose is a reducing sugar.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

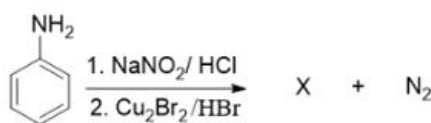
2. 2

3. 3

4. 4

Q.26

The product (X) in the following reaction is



- 1.
- 2.
- 3.
- 4.

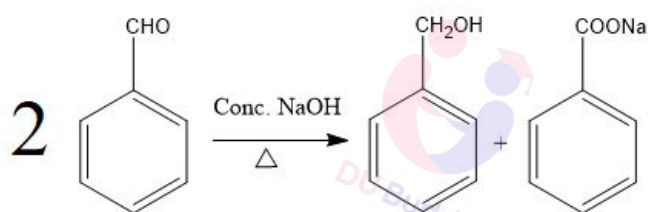
Options 1. 1

2. 2

3. 3

Q.27

The following reaction is an example of



1. Cannizzaro reaction
2. Hell-Volhard Zelinsky reaction
3. Friedel-Craft reaction
4. Gatterman-Koch reaction

Options 1. 1

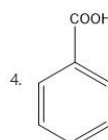
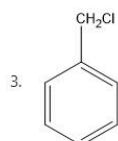
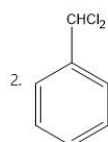
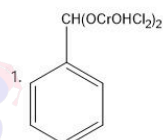
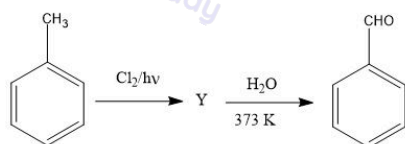
2. 2

3. 3

4. 4

Q.28

Identify the intermediate (Y) in the following reaction for the commercial method of manufacturing of benzaldehyde?



Options 1. 1

2. 2

3. 3

4. 4

Q.29 Match the complex species given in List-I with the possible isomerism given in List-II :

List-I  
Complex species

(A)  $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$

(B)  $\text{cis-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$

(C)  $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$

List-II  
Isomerism

(I) optical

(II) ionization

(III) coordination

(D)  $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$  (IV) geometrical

Choose the correct answer from the options given below:

1. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
2. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
3. (A)-(IV), (B)-(I), (C)-(III), (D)-(II)
4. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Options 1. 1

2. 2

3. 3

4. 4

Q.30 During denaturation of proteins which structure of proteins remain intact ?

1. Primary Structure
2. Secondary Structure
3. Tertiary Structure
4. Quaternary Structure

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Interstitial transition metal compounds

- (A) have higher melting points than those of pure metals.  
(B) have more reactivity than those of pure metals.  
(C) are non-stoichiometric compounds.  
(D) formed when small atoms like H, C or N are trapped inside the crystal lattice of the metals.

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (A), (B) and (C) only
3. (A), (B), and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.32 The magnetic moment in transition metals is calculated using the formula

1.  $\mu = \sqrt{n(n+2)}$

2.  $\mu = n\sqrt{(n+2)}$

3.  $\mu = g\sqrt{J(J+1)}$

4.  $\mu = \sqrt{J(J+1)}$

Options 1. 1

2. 2

3. 3

4. 4

Q.33 A coordination compound  $\text{CrCl}_3 \cdot 4\text{H}_2\text{O}$  precipitates silver chloride when treated with silver nitrate.

The molar conductance of its solution corresponds to a total of two ions.

What is the formula of the compound?

1.  $[\text{Cr}(\text{H}_2\text{O})_4]\text{Cl}_3$
2.  $[\text{CrCl}_3] \cdot 4\text{H}_2\text{O}$
3.  $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot \text{H}_2\text{O}$
4.  $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}$

Options 1. 1

2. 2

3. 3

4. 4

Q.34 Common oxidation state of lanthanoids is

1. +2
2. +3
3. +5
4. +4

Options 1. 1

2. 2

3. 3

4. 4

Match List-I (Name) with List-II (Structure)

| List-I              | List-II     |
|---------------------|-------------|
| (Name)              | (Structure) |
| (A) Cinnamaldehyde  | (I)         |
| (B) Vanillin        | (II)        |
| (C) Salicylaldehyde | (III)       |
| (D) Benzophenone    | (IV)        |

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

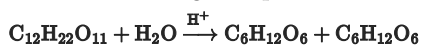
Options 1. 1

2. 2

3. 3

4. 4

Q.36 Inversion of cane sugar is a pseudo first order reaction



Choose the correct reason from the statements given below

- $\text{H}_2\text{O}$  is not involved in the reaction
- The concentration of water does not get altered during the reaction so the reaction behaves as a first order reaction
- Order of the reaction with respect to  $\text{H}_2\text{O}$  is zero
- The rate law equation for the reaction is  $k[\text{C}_{12}\text{H}_{22}\text{O}_{11}]^{1/2}[\text{H}_2\text{O}]^{1/2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Low level of Thyroxine hormone is correlated with

- Iodine deficiency
- Vitamin A deficiency
- Vitamin C deficiency
- Vitamin D deficiency

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Which among the following are coordination complexes?

- (A) Mohr's salt
- (B) Tetraamminediaquacobalt (II) ion
- (C) Carnallite
- (D) Potassium ferricyanide

Choose the correct answer from the options given below:

- 1. (A), (C) and (D) only
- 2. (A) and (C) only
- 3. (B) only
- 4. (B) and (D) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.39 Based on their rate laws, arrange the following reactions in the decreasing sequence of the reaction order.

- (A)  $\text{Rate} = k[\text{A}]^{3/2}[\text{B}]^{-1}$
- (B)  $\text{Rate} = k[\text{CHCl}_3][\text{Cl}_2]^{1/2}$
- (C)  $\text{Rate} = k[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]^0$
- (D)  $\text{Rate} = k[\text{A}]^{1/2}[\text{B}]^{3/2}$

Choose the correct answer from the options given below:

- 1. (A), (C), (B), (D)
- 2. (D), (C), (B), (A)
- 3. (D), (B), (C), (A)
- 4. (B), (C), (A), (D)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.40 The unit of rate constant for first order reaction is

- 1.  $\text{mol L}^{-1} \text{s}^{-1}$
- 2.  $\text{mol}^{-1} \text{L s}^{-1}$
- 3.  $\text{mol s}^{-1}$
- 4.  $\text{s}^{-1}$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.41 Read the passage carefully and answer the Questions

A homogeneous mixture of two or more than two components has uniform composition and properties. The binary solutions have two components, which may be solid, liquid or gas. The resultant solution has a larger amount of solvent and a smaller amount of solute. The amount of solute in a given amount of solvent decides the concentration of the resulting solution. The concentration can be expressed in various ways, like molarity, molality, normality and mole fraction.

The molality of a 1 liter aqueous solution of 93%  $\text{H}_2\text{SO}_4$  (mass/volume) is:

(Given: density of the solution =  $1.84 \text{ g mL}^{-1}$ , molar mass of  $\text{H}_2\text{SO}_4 = 98 \text{ g mol}^{-1}$ )

- 1.  $10.43 \text{ mol kg}^{-1}$
- 2.  $9.49 \text{ mol kg}^{-1}$
- 3.  $18.4 \text{ mol kg}^{-1}$
- 4.  $8.4 \text{ mol kg}^{-1}$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.42 Read the passage carefully and answer the Questions

A homogeneous mixture of two or more than two components has uniform composition and properties. The binary solutions have two components, which may be solid, liquid or gas. The resultant solution has a larger amount of solvent and a smaller amount of solute. The amount of solute in a given amount of solvent decides the concentration of the resulting solution. The concentration can be expressed in various ways, like molarity, molality, normality and mole fraction.

Glucose dissolved in water and ethanol dissolved in water are examples of

- 1. Liquid solution and liquid solution, respectively.
- 2. Solid solution and liquid solution, respectively.
- 3. Solid solution and solid solution, respectively.
- 4. Liquid solution and solid solution, respectively.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

**Q.43** Read the passage carefully and answer the Questions

A homogeneous mixture of two or more than two components has uniform composition and properties. The binary solutions have two components, which may be solid, liquid or gas. The resultant solution has a larger amount of solvent and a smaller amount of solute. The amount of solute in a given amount of solvent decides the concentration of the resulting solution. The concentration can be expressed in various ways, like molarity, molality, normality and mole fraction.

Molarity of a solution containing 8 grams NaOH in one liter of water is

1. 0.25 mol/L
2. 0.40 mol/L
3. 0.50 mol/L
4. 0.20 mol/L

Options 1. 1

2. 2
3. 3
4. 4

**Q.44** Read the passage carefully and answer the Questions

A homogeneous mixture of two or more than two components has uniform composition and properties. The binary solutions have two components, which may be solid, liquid or gas. The resultant solution has a larger amount of solvent and a smaller amount of solute. The amount of solute in a given amount of solvent decides the concentration of the resulting solution. The concentration can be expressed in various ways, like molarity, molality, normality and mole fraction.

Parts per million (ppm) is represented as

1. parts per million =  $\frac{\text{Number of parts of the component}}{\text{Total number of parts of all components of the solution}} \times 10^6$
2. parts per million =  $\frac{\text{Number of parts of the component}}{\text{Total number of parts of all components of the solution}} \times 10^5$
3. parts per million =  $\frac{\text{Number of parts of the component}}{\text{Total number of parts of all components of the solution}} \times 10^3$
4. parts per million =  $\frac{\text{Total number of parts of all components of the solution}}{\text{Number of parts of the component}} \times 10^6$

Options 1. 1

2. 2
3. 3
4. 4

**Q.45** Read the passage carefully and answer the Questions

A homogeneous mixture of two or more than two components has uniform composition and properties. The binary solutions have two components, which may be solid, liquid or gas. The resultant solution has a larger amount of solvent and a smaller amount of solute. The amount of solute in a given amount of solvent decides the concentration of the resulting solution. The concentration can be expressed in various ways, like molarity, molality, normality and mole fraction.

Which of the following property is independent of temperature?

1. Density
2. Mass
3. Molarity
4. Normality

Options 1. 1

2. 2
3. 3
4. 4

**Q.46** Read the passage carefully and answer the Questions

Alcohols and phenols are obtained when a hydrogen atom in aliphatic and aromatic hydrocarbons respectively, is replaced by -OH group. These compounds find wide applications in industry as well as in our daily life. You may notice that ordinary spirit is used for polishing wooden furniture which is chiefly a compound containing hydroxyl group. An alcohol contains one or more hydroxyl (OH) group(s) directly attached to carbon atom(s), of an aliphatic system (CH<sub>3</sub>OH) while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system (C<sub>6</sub>H<sub>5</sub>OH). Alcohols are prepared by the hydration of alkenes, fermentation of glucose, and reduction of carbonyl compounds. Alcohols produce alkenes upon dehydration whereas alkenes on acid catalyzed hydration convert into alcohols.

The product of the following reaction is



1.  $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}=\text{CH}_2$
2.  $\text{H}_3\text{C}-\underset{\text{H}}{\text{C}}=\underset{\text{H}}{\text{C}}-\text{OH}$
3.  $\text{H}_3\text{C}-\underset{\text{OH}}{\text{C}}-\text{CH}_3$
4.  $\text{HO}-\overset{\text{H}_2}{\text{C}}-\underset{\text{H}}{\text{C}}=\text{CH}_2$

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Read the passage carefully and answer the Questions

Alcohols and phenols are obtained when a hydrogen atom in aliphatic and aromatic hydrocarbons respectively, is replaced by -OH group. These compounds find wide applications in industry as well as in our daily life. You may notice that ordinary spirit is used for polishing wooden furniture which is chiefly a compound containing hydroxyl group. An alcohol contains one or more hydroxyl (OH) group(s) directly attached to carbon atom(s), of an aliphatic system ( $\text{CH}_3\text{OH}$ ) while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system ( $\text{C}_6\text{H}_5\text{OH}$ ). Alcohols are prepared by the hydration of alkenes, fermentation of glucose, and reduction of carbonyl compounds. Alcohols produce alkenes upon dehydration whereas alkenes on acid catalyzed hydration convert into alcohols.

Chemical formula of spirit used in polishing wooden furniture is

1.  $\text{CH}_3\text{OH}$

2.  $\text{C}_2\text{H}_5\text{OH}$

Extra close brace or missing open brace

4.  $(\text{CH}_3)_3\text{COH}$

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the passage carefully and answer the Questions

Alcohols and phenols are obtained when a hydrogen atom in aliphatic and aromatic hydrocarbons respectively, is replaced by -OH group. These compounds find wide applications in industry as well as in our daily life. You may notice that ordinary spirit is used for polishing wooden furniture which is chiefly a compound containing hydroxyl group. An alcohol contains one or more hydroxyl (OH) group(s) directly attached to carbon atom(s), of an aliphatic system ( $\text{CH}_3\text{OH}$ ) while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system ( $\text{C}_6\text{H}_5\text{OH}$ ). Alcohols are prepared by the hydration of alkenes, fermentation of glucose, and reduction of carbonyl compounds. Alcohols produce alkenes upon dehydration whereas alkenes on acid catalyzed hydration convert into alcohols.

An alkyl halide can be converted into alcohol through

1. addition

2. elimination

3. substitution

4. dehydrohalogenation

Options 1. 1

2. 2

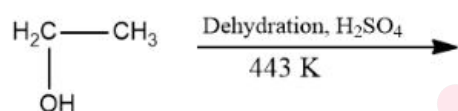
3. 3

4. 4

Q.49 Read the passage carefully and answer the Questions

Alcohols and phenols are obtained when a hydrogen atom in aliphatic and aromatic hydrocarbons respectively, is replaced by -OH group. These compounds find wide applications in industry as well as in our daily life. You may notice that ordinary spirit is used for polishing wooden furniture which is chiefly a compound containing hydroxyl group. An alcohol contains one or more hydroxyl (OH) group(s) directly attached to carbon atom(s), of an aliphatic system ( $\text{CH}_3\text{OH}$ ) while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system ( $\text{C}_6\text{H}_5\text{OH}$ ). Alcohols are prepared by the hydration of alkenes, fermentation of glucose, and reduction of carbonyl compounds. Alcohols produce alkenes upon dehydration whereas alkenes on acid catalyzed hydration convert into alcohols.

Product of the following reaction is



1.  $\begin{array}{c} \text{H}_3\text{C}-\text{C}=\text{CH}_2 \\ | \\ \text{OH} \end{array}$
2.  $\text{H}_2\text{C}=\text{CH}_2$
3.  $\text{HO}-\text{C}(\text{H})=\text{C}(\text{H})-\text{OH}$
4.  $\text{HO}-\text{C}(\text{H}_2)-\text{C}(\text{H})=\text{CH}_2$

Options 1. 1

2. 2

3. 3

4. 4

**Q.50** Read the passage carefully and answer the Questions

Alcohols and phenols are obtained when a hydrogen atom in aliphatic and aromatic hydrocarbons respectively, is replaced by -OH group. These compounds find wide applications in industry as well as in our daily life. You may notice that ordinary spirit is used for polishing wooden furniture which is chiefly a compound containing hydroxyl group. An alcohol contains one or more hydroxyl (OH) group(s) directly attached to carbon atom(s), of an aliphatic system ( $\text{CH}_3\text{OH}$ ) while a phenol contains -OH group(s) directly attached to carbon atom(s) of an aromatic system ( $\text{C}_6\text{H}_5\text{OH}$ ). Alcohols are prepared by the hydration of alkenes, fermentation of glucose, and reduction of carbonyl compounds. Alcohols produce alkenes upon dehydration whereas alkenes on acid catalyzed hydration convert into alcohols.

Which of the following compounds lacks phenolic group?

1. Quinol
2. Cyclohexanol
3. Catechol
4. Picric acid

Options 1. 1

2. 2

3. 3

4. 4